

Upgrading of pinyon-juniper catalytic pyrolysis oil via hydrodeoxygenation

Hossein Jahromi^{*}, Foster A. Agblevor

USTAR Bioenergy Center, Department of Biological Engineering, Utah State University, UT 84322, USA



ARTICLE INFO

Article history:

Received 10 October 2016

Received in revised form

17 November 2017

Accepted 26 November 2017

Available online 27 November 2017

ABSTRACT

In this study we discuss hydrodeoxygenation (HDO) of pinyon juniper (PJ) catalytic pyrolysis oil over Ni/SiO₂–Al₂O₃ catalyst in a batch reactor to improve the physicochemical properties of the oil. The influence of temperature (350–500 °C), reaction time (15–90 min), and initial hydrogen pressure (3.5–10 MPa), on hydrodeoxygenation of PJ pyrolysis oil was investigated. After hydrogenation was completed, gas, coke, and a liquid product of two immiscible phases (aqueous and organic), were obtained. Maximum HDO of bio-oil was achieved at 450 °C while the initial hydrogen pressure was 7 MPa and the reaction time was 30 min. Under these conditions, the H/C and O/C atomic ratios changed from 1.29 to 0.29 respectively for bio-oil to 2.36 and 0 for HDO oil respectively. The higher heating value increased from 27.64 MJ/kg of bio-oil to 45.58 MJ/kg of upgraded oil. The water content of organic liquid product was less than 0.05 wt% while it was 1.63 wt% in the feed. The viscosity of upgraded oil was 1.26 cP compared to 119 cP for the crude bio-oil.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

The global demand for energy is growing due to the development of society. Considering the limited supply of fossil fuels, the need for sustainable renewable fuels is increasing [1]. Fast pyrolysis technique has drawn scientists' attention in recent decades because of its potential to produce liquid fuels. Fast pyrolysis is a thermochemical decomposition process that occurs in absence of oxidizing agents at elevated temperatures [2–5]. This process is highly effective at breaking down macromolecular structures into smaller organic compounds [3]. A number of studies on fast pyrolysis oils of different types of feedstock such as woody and proteinaceous biomasses, grasses, and animal waste have indicated the different properties of the final bio-oil [4–8]. However, their qualities were not suitable for direct refinery due to poor stability, high acidity, and low heating value [9–12].

The high oxygen and water content of bio-oil lowers its heating value. The pH of bio-oil is usually between 2 and 4 due to the presence of carboxylic acids [13,14]. The acidic nature of bio-oil causes harsh conditions for the equipment used for transportation, storage, and processing. Another problem that is

associated with bio-oil is stability during storage. The instability of bio-oil is attributed to presence of highly reactive organic compounds causing repolymerization. Moreover, organic acids, aldehydes, and ketones can react to form hemiacetals, acetals, and ethers respectively [15,16]. Overall the unfavorable properties of pyrolysis oils make it impossible to use as drop-in transportation fuels [17–19].

Intensive studies have been carried out on upgrading of pyrolysis oils through different techniques such as esterification under supercritical ethanol, hydrogenation/hydrodeoxygenation (HDO), hydrocracking, emulsification, solvent addition, and steam reforming [17,19,20]. Among these techniques, catalytic hydrodeoxygenation of pyrolysis oil is one of the most efficient technologies to improve its fuel quality. In addition to hydrogenation of unsaturated compounds, the majority of hydrogenation catalysts normally favor several other reactions simultaneously such as cracking, decarbonylation, decarboxylation, and hydrodeoxygenation [11,20,21].

Production of value-added hydrocarbons is one of the major goals of bio-oil upgrading. Aromatic production via HDO is desirable, as this indicates that the catalyst(s) deoxygenate the oil more efficiently with less hydrogenation, resulting in less hydrogen consumption. Moreover, one-ring aromatic compounds are valuable chemicals. An investigation of production of aromatics through HDO of model bio-oil over mesoporous Ni/SBA-15 and Co/

^{*} Corresponding author.

E-mail address: hossein.jahromi@aggiemail.usu.edu (H. Jahromi).

SBA-15 has been shown that Ni/SBA-15 catalyst exhibit much higher catalytic performance as compared with Co/SBA-15 catalyst for the hydrodeoxygenation reaction performance of anisole. While the Ni/SBA-15 catalyst would lead to the appearance of partial hydrodeoxygenation products due to the hydrogenation ability of nickel active phase, Co/SBA-15 could selectively hydrogenolysis the C–O bond and could act as the potential hydrodeoxygenation catalyst for aromatic production from the bio-oils [22]. In the case of bio-oil, it has been reported that a first stage HDO at temperatures up to 270 °C is required to convert highly reactive compounds such as aldehydes and ketones so that second stage HDO could be conducted at higher temperatures for complete removal of oxygen [23]. Most HDO studies have been performed using sulfided NiMo and CoMo as suitable catalysts. Moreover, noble metal catalysts such as Pt/C, Ru/C, Pd/C, and Ru/TiO₂ have also been tested and shown to result in higher levels of hydrodeoxygenation than the traditional HDO catalysts such as sulfided CoMo [24–26]. However noble metal catalysts are relatively more expensive than common HDO catalysts. The influence of catalyst support on HDO activity and stability of catalyst has also been investigated. Ni and Ru supported on silica gel, nano spring (NS) silica, and Al₂O₃ have been tested for HDO of phenol as a model compound [27]. Silica based Ni catalysts performed better than alumina based. The Ni(20%)-NS was the best performing Ni catalyst in terms of deoxygenation. Of the Ru based catalysts, the Ru-NS were the best performing, exhibiting nearly complete phenol conversion and deoxygenation [27].

Although various studies have been published on HDO of different types of biomass pyrolysis oils, the majority of previous research on bio-oil HDO was generally carried out to evaluate the effect of catalysts [28,29] thus, parametric studies require more detailed investigations to find the optimum feedstock and operating conditions for bio-oil hydrotreating to obtain high quality hydrocarbons.

According to the U.S. Billion Ton update [30], biomass for bio-energy would come from a variety of sources such as agricultural crops (i.e. corn stover), herbaceous energy crops (i.e. switchgrass), forests (i.e. pine bark), and short rotation woody biomass (i.e. pine, hybrid poplar, and pinyon juniper (PJ)). PJ woodlands occupy about 30 million hectares in the western United States (including states of Arizona, California, Colorado, Idaho, Utah, Nevada, New Mexico, Oregon, and Wyoming). Domination of PJ woodlands decreases the herbaceous vegetation and increases bare lands, thus causes soil erosion and nutrition loss [4]. Studies have shown that expansion of PJ woodlands has reduced the amount of precipitation, increased soil erosion, and increase the potential of crown fires. Hence, land management agencies are aiming on reducing the population of PJ woodlands by bulldozing, chaining, hand cutting, mechanical removal, and prescribed fire [31–34]. Therefore, PJ can be a potential biomass feedstock for production of bio-oil.

The objective of this research is to study hydrogenation/deoxygenation of PJ catalytic pyrolysis oil over commercial Ni/SiO₂–Al₂O₃ catalyst. Unlike traditional HDO catalysts such as CoMo/Al₂O₃ and NiMo/Al₂O₃, this catalyst does not require pre-sulfidation. Furthermore, Ni/SiO₂–Al₂O₃ is cheaper than precious metal HDO catalysts such as Pt, Pd, and Ru [24,26,35]. In this study we discuss the effect of temperature, hydrogen pressure, and reaction time on the HDO of PJ pyrolysis oil.

2. Material and methods

2.1. Material

PJ biomass chips were supplied by the U.S. Bureau of Land Management. Red Mud, supplied by Almatris (Burnside, LA, USA)

was used as catalyst for fast pyrolysis of biomass. The wet red mud was dried at room temperature and then ground and sieved to a particle size of 125–180 µm for fluidized-bed pyrolysis. The ground particles were calcined at 550 °C in a muffle furnace (Thermo Scientific, Inc., Waltham, MA) for 5 h before being used for the pyrolysis. The detailed characterization of the red mud has been reported by Yathavan and Agblevor [4]. Commercial Nickel on silica/alumina (~65 wt % loading Ni) catalyst powder, obtained from Sigma-Aldrich (St. Louis, MO, USA), was used (as received) in HDO experiments. High purity (99%) hydrogen (Airgas, PA, USA) was used for HDO experiments.

2.2. Pyrolysis of biomass

PJ chips ground to pass a 2-mm sieve, were used as a raw material for production of catalytic pyrolysis oil. Catalytic pyrolysis oil was produced at 450 °C using a pilot scale fluidized bed reactor system, the details of which can be found elsewhere [36]. The pyrolytic products were condensed to liquid using a series of two water-cooled condensers and an electrostatic precipitator (ESP). The oil product that was collected by ESP (operating at 30 kV) was used for HDO studies.

2.3. Hydrodeoxygenation of bio-oil

The bio-oil was subjected to HDO reactions in a Parr Series 4571 1000 mL autoclave reactor (Parr Instruments, Moline, IL, USA). The reactor can withstand a maximum pressure of 35 MPa at 500 °C. A Parr 4848 controller was used to control the internal temperature and impeller speed. In a typical test, bio-oil (30 g) and catalyst (6 g) were loaded into the reactor. The reactor was first flushed with nitrogen three times and then flushed with hydrogen three times to purge the reactor. High purity hydrogen was supplied from a reservoir tank via a pressure regulator. The reactor was then pressurized with hydrogen (3.5, 5.2, 7, 8.6, or 10 MPa) and a gas sample was taken from a gas release valve from the gas sampling port for gas analysis when the reactor was at room temperature. The reactor was then heated to reaction temperature (350, 400, 450, or 500 °C) at heating rate of 10 °C/min using a heating mantle. The reaction time was recorded when the required temperature was reached. The stirrer speed was kept constant (~1000 rpm) in all experiments.

After the desired reaction time (15, 30, 60, or 90 min), the reactor was cooled to room temperature using the internal cooling coil and an external air fan simultaneously. A gas sample was collected in a tedlar bag for gas analysis when the reactor was cooled down to room temperature. Hydrogen consumption was measured using equation (1).

$$H_2 \text{ consumption} \left(\frac{\text{mole } H_2}{\text{kg biooil}} \right) = \left(n_{i_{H_2}} - x_{f_{H_2}} \cdot n_{f_{tot}} \right) \times \frac{1}{30 \text{ g biooil}} \times \frac{1000 \text{ g}}{1 \text{ kg}} \quad (1)$$

where $n_{i_{H_2}}$ is the initial number of mole of hydrogen, $x_{f_{H_2}}$ is the final mole fraction of hydrogen, and $n_{f_{tot}}$ is the total number of mole of gas at the end of experiment.

After each experiment, the liquid products were collected in centrifuge test tubes and centrifuged for 30 min at g-force of 2147 to separate the resulting aqueous and organic phases and residual solids and catalyst. Both liquid phases were separated and weighed for mass balance computation. The solids (catalyst and coke) were collected and dried at 95 °C for 6 h. The vessel and reactor parts were rinsed with methanol-toluene mixture (1:1 vol ratio) to

collect any remaining catalyst and oil. The solvent washings were filtered through Watman 42 ashless filter paper (GE Healthcare, UK) and dried at 95 °C. The weight of filter paper was recorded before and after filtration.

The total mass of gaseous product was calculated back using equation (2):

$$W_g = \sum_i x_i \cdot MW_i \cdot n_{tot} \quad (2)$$

where W_g is the total mass of gaseous product (g), x_i is the mole fraction of gas i , MW_i is the molecular weight of gas i (g/mole), and n_{tot} is the total number of moles of gas product. The yield of liquid, gas, and solid product were calculated using equations (3)–(5) respectively.

$$Y_{liquid}(\%) = \frac{W_l}{W_f} \times 100 \quad (3)$$

$$Y_{gas}(\%) = \frac{W_g}{W_f} \times 100 \quad (4)$$

$$Y_{coke}(\%) = \frac{(W_s - W_c)}{W_f} \times 100 \quad (5)$$

where W_f is the mass of feed bio-oil and consumed hydrogen combined (g), W_l is the mass of liquid product (g), W_g is the mass of gas product (g), W_s is the weight of total solid residues (g), and W_c is the weight of catalyst (g).

An elemental carbon balance before and after each experiment was performed.

In a blank experiment, without catalyst, 30 g of PJ bio-oil was charged into the reactor and the reactor was pressurized to 7 MPa with hydrogen. The bio-oil was allowed to react for 60 min at 450 °C. All experiments were conducted in duplicate.

2.4. Characterization of PJ wood, catalytic pyrolysis oil, and upgraded oil

2.4.1. Physical properties

The moisture content of the PJ wood was determined according to standard method ASTM E1756- 08 [37]. The ash content of the PJ wood was determined according to ASTM E1755-01 method [38]. The water content of the crude bio-oil and hydrotreated oil were determined by Karl-Fisher titration method with Hydranal®-composite 5 solution. A Metrohm 701KF Titrino and 703 titration stand setup (Brinkmann Instruments, Riverview, FL, USA) were used for the volumetric Karl Fischer titration. The pH was measured using Mettler Toledo pH Meter and probe (Mettler–Toledo GmbH, Switzerland). The density and kinematic viscosity of the bio-oil and hydrogenated oil were measured at 40 °C using Anton Parr Stabinger viscometer svm 3000 (Ashland, VA, USA).

2.4.2. Chemical properties

The higher heating value (HHV) of PJ wood, bio-oil, and hydrotreated oil were determined using an IKA Model C2000 basic bomb calorimeter (Wilmington, NC, USA). The elemental composition of biomass, bio-oil, and hydrogenated oil were determined using ThermoFischer Scientific Flash 2000 organic elemental analyzer (ThermoFischer Scientific, Waltham, MA, USA). Fourier transform infrared spectroscopy (FT-IR) over the range of 600–4000 nm was recorded using Avatar 360 FT-IR instrument (ThermoFischer Scientific, Waltham, MA, USA). The ^{13}C NMR spectra were recorded on a JEOL 300 MHz NMR spectrometer (JEOL Ltd.,

Japan). In a 5-mm sample probe about 0.5 g of oil was dissolved in 1.5 g deuterated solvent. Dimethyl sulfoxide- d_6 (DMSO- d_6) (Sigma-Aldrich, St. Louis, MO, USA) was used for catalytic pyrolysis oil and chloroform- d (Cambridge Isotope Laboratories, Inc., USA) was used for HDO oil. The observing frequency for the ^{13}C nucleus was 100.58 MHz, the pulse width was 10 μs , the acquisition time was 1.58 s, and the relaxation delay was 2 s. The spectra were obtained with 3000 scans and a sweep width of 20 kHz.

2.5. Gas analysis

Gas samples were collected in tedlar sampling bags from the sampling port installed on the reactor head and injected to a Varian 490- micro GC system (Agilent Technology, Santa Clara, CA, USA). The micro GC was equipped with two modules: a 10 m Molsieve 5A (MS) column, and a 10 m porous polymer (PPU) column. Each module had a thermal conductivity detector. MS column was used to analyze hydrogen, methane, and carbon monoxide (CO), while carbon dioxide (CO_2) and C1–C5 hydrocarbons were analyzed on the PPU column. Argon and helium were used as carrier gases for MS column and PPU column respectively. Gas concentrations were calculated relative to calibration curves of three standard gas mixtures supplied from Scotty Specialty Gases (Fremont, CA, USA). Gas samples were analyzed for H_2 , CO, CO_2 , CH_4 , C_2H_4 , C_2H_6 , C_3H_8 , C_4H_{10} , and C_5H_{12} .

3. Results and discussion

3.1. Characterization of PJ wood and catalytic pyrolysis oil

The biomass was characterized on the basis of moisture, ash content, elemental composition, and higher heating value (HHV). The physicochemical properties of the PJ wood are shown in Table 1. The ESP-captured oil was used in HDO experiments. The ESP oil were characterized based on ash content, elemental composition, HHV, pH, water content, density, and dynamic viscosity. The yield of organic liquid after fast pyrolysis of biomass was about 30% mass of fed PJ dry basis. The physicochemical properties of the raw bio-oil are shown in Table 1.

3.2. HDO experiments results

3.2.1. Influence of reaction temperature

3.2.1.1. Mass balance of HDO products. The HDO of bio-oil was performed in the temperature range of 350–500 °C. In all of the experiments the catalyst to feed mass ratio was 0.2, the reaction time was 60 min, and the initial hydrogen pressure was 7 MPa. The

Table 1
Characterization of Pinyon-Juniper biomass and bio-oil^a.

Properties	PJ biomass	Catalytic PJ bio-oil
Composition (wt%)		
Ash content	0.53 ± 0.06	0.00
Nitrogen	0.17 ± 0.01	0.47 ± 0.04
Carbon	54.43 ± 0.11	66.88 ± 0.21
Hydrogen	6.27 ± 0.09	7.20 ± 0.07
Sulfur	0.00	0.00
Oxygen ^b	38.60 ± 0.12	25.42 ± 0.12
HHV (MJ/kg)	19.37 ± 0.11	27.64 ± 0.23
pH	NA ^c	3.46 ± 0.05
Water content (wt%)	6.65 ± 0.09	1.63 ± 0.08
Density (g/cm ³)	0.55 ± 0.03	1.16 ± 0.02
Dynamic viscosity (cP)	NA ^c	119 ± 5

^a Error are the standard deviation of two measurements.

^b By difference.

^c Not applicable.

products yield distribution vs. temperature is shown in Fig. 1.

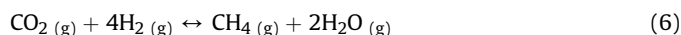
As expected the reaction temperature influenced the product yields. As the reaction temperature increased from 350 °C to 450 °C, the yield of organic phase decreased from 55.7% to 44.8% and the gas yield increased from 21.1% to 27.6%, while the yield of aqueous phase slightly increased from 19.4% to 21.1%. Also the coke yield increased from 2.6% to 5.3% over the temperature range of 350–450 °C. However, by increasing the temperature from 450 °C to 500 °C, the organic liquid yield decreased dramatically from 44.8% to 0% while the gas yield increased considerably from 27.6% to 70.4%. This phenomena can be explained by complete cracking of bio-oil hydrocarbons into gas at 500 °C. By increasing the temperature, the oil was converted to gas rather than increased hydrodeoxygenation of the oil [39]. By increasing the temperature from 450 to 500 °C, the aqueous phase yield decreased slightly from 21.1% to 18.2%. This result is explained in section 3.2.1.2. The total mass balance and carbon balance of all HDO experiments at different reaction temperatures were between 98 and 99% (Table 2). (Pressure profile during HDO experiments at different temperatures is demonstrated in Fig. S1 in supplementary data).

3.2.1.2. Characterization of HDO products. Reaction temperature significantly influenced the physicochemical properties of HDO oil. Table 3 illustrate the properties of HDO oil and gas product analysis at different temperatures. Increasing reaction temperature from 350 °C to 450 °C increased the hydrogen content of HDO oil from 10.42 wt.% to 16.41 wt.% while the oxygen content were remarkably reduced from 20.12 wt.% to 0 wt.%. This would indicate the increased hydrodeoxygenation of the oil over this temperature range. Hydrogen consumption increased gradually from 48 mol/kg to 57 mol/kg in this temperature range. The density of oil reduced by 10.23% from 0.88 g/cm³ to 0.79 g/cm³ while the temperature increased from 350 °C to 450 °C. Also the dynamic viscosity decreased from 3.22 cP to 1.35 cP over this temperature range. The reduction in density and viscosity could be attributed to hydrocracking of high molecular weight compounds in the bio-oil during HDO [40]. The reduction in acidity can be due to conversion of carboxylic acids in the bio-oil during hydrodeoxygenation [10].

At 350 °C the major gas product was methane (94.41%) and small amounts of CO₂ and C₂–C₅ hydrocarbon gasses were formed (Table 3). When the temperature was increased from 350 °C to 450 °C, the amount of C₂–C₅ hydrocarbon gases increased indicating further cracking of the bio-oil compounds [39]. Also no CO₂ was detected at 400–450 °C. This would suggest the methanation of carbon dioxide over nickel (known as Sabatier reaction). This

reaction takes place between 250 °C and 500 °C [41].

Increasing the reaction temperature from 450 °C to 500 °C resulted in high consumption of hydrogen (91 mol/kg). As previously mentioned (Fig. 1), overcracking of hydrocarbon molecules at 500 °C resulted in complete conversion of oil into gas products. The major gas component was methane (84.72%) at this temperature. This would suggest that at 500 °C hydrocracking of the bio-oil compounds result in production of methane that would confirm the high amount of hydrogen consumption. At this temperature, 3.10% CO₂ was detected in the gas product. Since methanation of CO₂ is a reversible reaction [41], this would explain the reduction of aqueous liquid yield from 21.1% to 18.2% by increasing the temperature from 450 °C to 500 °C (Fig. 1). At 500 °C overcracking of hydrocarbon compounds resulted in formation of high amounts of methane. Methane could then partially react with the water produced from HDO and form CO₂ and hydrogen (6).



3.2.2. Influence of hydrogen pressure

3.2.2.1. Mass balance of HDO products. A set of experiments was carried out in the range of 3.5–10 MPa of initial hydrogen pressure. In these experiments, reaction temperature and reaction time were kept constant at 450 °C and 60 min respectively. The data obtained on product yields distribution is shown in Fig. 2.

Increasing hydrogen pressure from 3.5 MPa to 7 MPa increased the organic liquid yield from 40.3% to 44.8%. This could be due to increase in concentration of adsorbed hydrogen on catalyst surface and increase in the amount of dissolved H₂ in the oil, as hydrogen pressure was increased [23,42]. Over this pressure range, the gas yield reduced from 32.3% to 27.6% and the coke yield decreased from 10.4% to 5.3%. This would suggest that higher levels of hydrogen pressure would help HDO reactions to predominate cracking and (re)polymerization reactions that produce gas and coke [43]. By increasing the initial hydrogen pressure from 3.5 MPa to 7 MPa, the yield of aqueous phase product increased from 15.7% to 21.1%. Since water is a product of HDO this would suggest that increasing the hydrogen pressure promotes the hydrodeoxygenation of bio-oil compounds.

Increasing the hydrogen pressure from 7 MPa to 10 MPa did not have a significant effect on product yields as is illustrated in Fig. 2. These results indicate that solubility of hydrogen in the oil and adsorbed hydrogen on the catalyst surface reached a maximum (saturated) value at 7 MPa [23,43]. Thus, increasing hydrogen pressure to higher levels would not have significant effect on HDO, cracking, and condensation reactions. Table S1 (in supplementary data) presents the total mass balance and carbon balance of all HDO experiments at different initial hydrogen pressures. Also for more information on pressure profile during HDO experiments at different hydrogen initial pressure refer to Fig. S2 in supplementary data.

3.2.2.2. Characterization of HDO products. Tables S2 (in supplementary data) shows the physicochemical properties of HDO oil and gas product analysis at different hydrogen pressures. By increasing H₂ pressure from 3.5 MPa to 7 MPa hydrogen content of HDO oil increased from 10.24 wt.% to 16.41 wt.% and its oxygen content considerably decreased from 18.23 wt.% to 0 wt.%. These results indicate that higher pressure of hydrogen favors higher level of both oil hydrogenation and deoxygenation. As previously mentioned, this could be due to the increased concentration of hydrogen in the liquid phase [23]. The water content of the HDO oil reduced from 1.07 wt.% to less than 0.05 wt.%. This could be due to

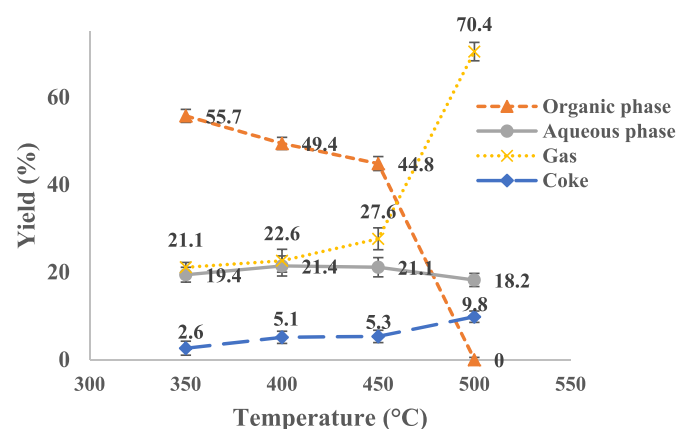


Fig. 1. Effect of temperature on HDO product yield distributions. The H₂ initial pressure, and reaction time were 7 MPa, and 60 min respectively.

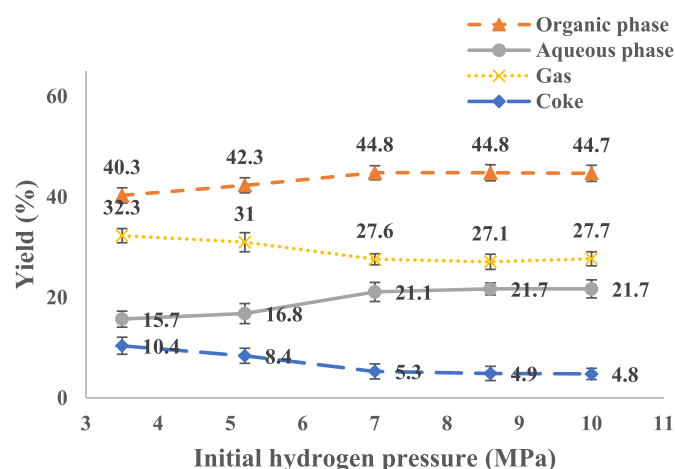
Table 2

Total mass balance and carbon mass balance of HDO experiments at different temperatures.

T (°C)	Total mass balance (wt.%)					Carbon balance (wt.%)				
	Organic	Aqueous	Gas	Coke	SUM	Organic	Aqueous	Gas	Coke	SUM
350	55.7	19.4	21.1	2.6	98.8	72.7	2.1	20.1	3.8	98.7
400	49.4	21.4	22.6	5.1	98.5	68.8	0.5	21.5	7.4	98.2
450	44.8	21.1	27.6	5.3	98.8	68.8	0	22.1	7.8	98.7
500	0	18.2	70.4	9.8	98.4	0	0	84.3	14.4	98.7

Table 3H₂ consumption, physicochemical properties of the liquid products, and gas product composition at different temperatures^b.

Properties	Reaction time and hydrogen initial pressure were 60 min and 7 MPa respectively			
	350 °C	400 °C	450 °C	500 °C
H ₂ consumption (mol/kg bio-oil)	48 ± 8	50 ± 7	57 ± 3	91 ± 2
Aqueous phase				
Water content (wt%)	93.23 ± 0.10	97.26 ± 0.15	99.51 ± 0.15	99.17 ± 0.25
pH	5.11 ± 0.12	6.34 ± 0.17	6.85 ± 0.05	6.87 ± 0.34
HDO oil (organic phase)				
Elemental analysis (wt%)				
N	0.21 ± 0.02	0.13 ± 0.01	0.12 ± 0.01	NA
C	69.25 ± 0.85	75.12 ± 1.14	83.47 ± 1.09	NA
H	10.42 ± 0.75	12.54 ± 0.65	16.41 ± 0.43	NA
O	20.12 ± 0.05	12.21 ± 0.06	0.00	NA
HHV (MJ/kg)	42.07 ± 0.67	42.57 ± 0.91	45.44 ± 0.84	NA
pH	5.72 ± 0.33	6.53 ± 0.21	NA	NA
Water content (wt%)	<DL ^a	<DL ^a	<DL ^a	NA
Density (g/ml)	0.88 ± 0.02	0.84 ± 0.02	0.79 ± 0.01	NA
Dynamic viscosity (cP)	3.22 ± 0.01	1.58 ± 0.01	1.35 ± 0.01	NA
Gas composition (mole %)				
H ₂	0.00	0.00	0.00	11.75 ± 0.22
CO ₂	1.33 ± 0.02	0.00	0.00	3.10 ± 0.05
CH ₄	94.41 ± 0.23	87.22 ± 0.71	85.83 ± 0.42	84.72 ± 0.81
C ₂ H ₄	2.73 ± 0.02	7.22 ± 0.10	9.02 ± 0.03	0.51 ± 0.01
C ₃ H ₈	1.04 ± 0.01	3.52 ± 0.08	3.82 ± 0.01	0.01
C ₄ H ₁₀	0.41 ± 0.01	1.57 ± 0.01	1.18 ± 0.01	0.01
C ₅ H ₁₂	0.05	0.46 ± 0.01	0.15	0.00

^a Errors are the standard deviation of two measurements.^b The detection limit was 0.05%.**Fig. 2.** Effect of H₂ pressure on HDO product yield distributions. The temperature and reaction time were 450 °C and 60 min respectively.

the movement of water from organic phase to aqueous phase as HDO reactions progress [44,45]. The heating value increased from 42.72 MJ/kg to 45.44 MJ/kg. Like previous experiments the major gas product was methane and a small amount of C₂–C₅ hydrocarbon gases were produced in all experiments. The concentration of C₂–C₅ gases increased by increasing the hydrogen initial pressure. This would suggest that hydrocracking of the bio-oil compounds

increased by increasing hydrogen pressure [46].

Increasing the pressure from 7 MPa to 10 MPa did not have a significant effect on the oil and gas composition. These results suggest that at hydrogen initial pressure of 7 MPa, dissolved hydrogen on the catalyst surface reached a maximum value and increasing the pressure to higher levels will not affect HDO reactions [24].

3.2.3. Influence of reaction time

3.2.3.1. Mass balance of HDO products. To examine the effect of reaction time on product yield distributions, HDO experiments were performed at four different reaction times. In all experiments the temperature and initial hydrogen pressure were kept at 450 °C and 7 MPa respectively. The reactor was maintained at 450 °C for 15, 30, 60, and 90 min and cooled down to room temperature afterwards. The yields of HDO products for these reaction times are illustrated in Fig. 3.

Increasing the reaction time from 15 to 90 min significantly decreased the organic liquid yield from 53.7% to 37.8%. The gas yield increased from 25.8% to 34.1% over time. This result implies that the organic compounds in the oil phase were converted to gas by further cracking rather than increased HDO of the oil [39]. From 15 to 60 min, the aqueous liquid yield slightly increased from 17.4% to 21.1% that indicates the hydrodeoxygenation reaction progress that removes oxygen in the form of water. Then it slightly decreased to 18.1% as the reaction time increased to 90 min. The latter observation is explained in section 3.2.3.2. Increasing the reaction time

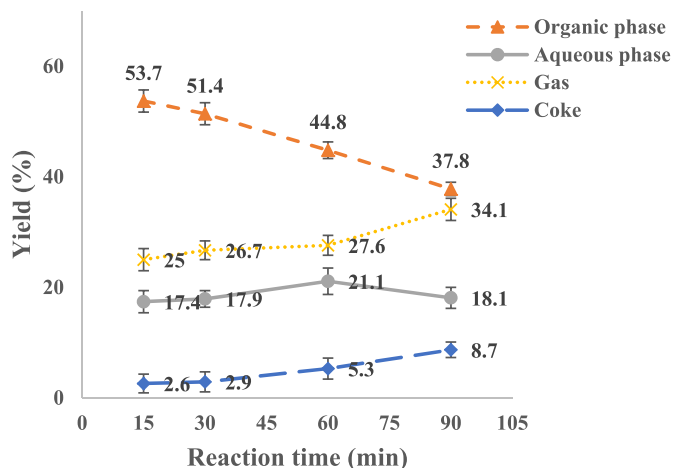


Fig. 3. Effect of reaction time on HDO products yield distributions. The temperature and H_2 initial pressure were 450 °C and 7 MPa respectively.

from 15 to 90 min increased the coke yield from 2.6% to 8.7%. It suggests that side reactions such as repolymerization and recombination might have occurred over longer reaction times, resulting in coke formation [10]. (For detailed information on total mass balance and carbon balance of HDO experiments at different reaction times refer to Table S3 in supplementary data. Also more information about pressure profile during HDO experiments at different reaction time is provided in Fig. S3 in supplementary data).

3.2.3.2. Characterization of HDO products. As shown in Table S4 (supplementary data), increasing the reaction time from 15 to 30 min increased the hydrogen content from 13.27 wt.% to 16.42 wt.% and decreased the oxygen content from 8.11 wt.% to 0 wt.%. Hydrogen consumption was slightly increased from 46 mol/kg to 50 mol/kg. In this reaction time range water content, density, and dynamic viscosity slightly improved.

Increasing the reaction time from 30 min to 60 and 90 min did not have a significant effect on elemental composition of HDO oil, HHV, water content, density, and viscosity. However, H_2

consumption increased from 50 mol/kg to 57 mol/kg and 60 mol/kg respectively (Table S4).

In all experiments at different reaction times, the major gas product was methane. At reaction time of 90 min, 1.75% CO_2 was detected in gas products that could be formed from reverse methanation reaction as discussed before [41]. This would also explain the reduction of aqueous phase yield while the reaction time was increased from 60 to 90 min (Fig. 3).

These results suggest that 30 min reaction time was long enough for completion of HDO reactions and longer reaction time result in further hydrocracking of the organic components and conversion of organic compounds to gas phase as previously shown in Fig. 3.

3.3. FT-IR and NMR analysis of organic liquid products

The FT-IR spectra of the raw bio-oil and HDO oils at different temperatures are presented in Fig. 4. The absorbance peaks between 3050 and 3650 cm^{-1} were ascribed to $-OH$ stretching vibration. This result indicated that phenolic compounds and alcohols were produced during catalytic pyrolysis of PJ biomass. These peaks were also ascribed to presence of water in the bio-oil. With the increase in temperature, these peaks gradually weakened or even disappeared in the HDO oil that could be due to phenolic or alcoholic hydroxyl cleavage, elimination, intramolecular dehydration, or hydrodeoxygenation of $-OH$ groups. Furthermore, during HDO, water could possibly migrate to the aqueous phase. The peaks between 2840 and 3010 cm^{-1} and the two other bands at 1376 and 1453 cm^{-1} were caused by $C-H$ stretching and deformation in methyl groups and the methylene groups, respectively. The peaks were intensified after HDO compared to the original bio-oil [47,48]. The peaks at 1590–1610 cm^{-1} were attributed to in-ring $C-C$ stretching. These peaks disappeared completely at 450 °C suggesting that the aromatics could possibly undergo cracking, or polymerization to form coke. The peak at 1670 cm^{-1} was ascribed to $C=O$ stretching, which gradually disappeared with the increase of temperature. The appearance of typical carbonyl group $C=O$ stretching vibrations at 1710 cm^{-1} showed that aldehydes, ketones, or carboxylic acids were only produced at 350 °C [49]. At higher temperatures the $C-O$ stretching intensity of absorption band at 1033 and 1100 cm^{-1} faded away, which could be described as the

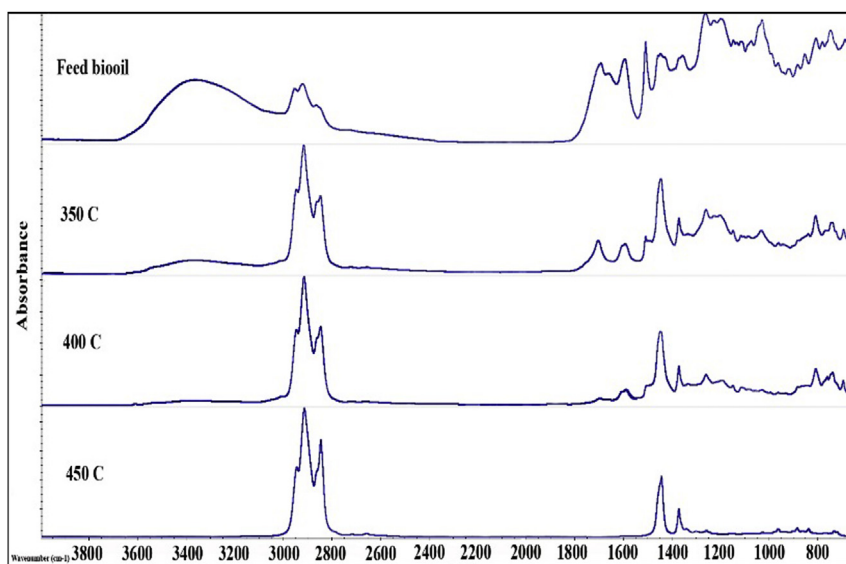


Fig. 4. FT-IR spectra of HDO oils obtained at different temperatures. H_2 initial pressure, and reaction time were 7 MPa and 60 min respectively.

deoxygenation of alcohols, phenols, or esters. Moreover, several bands that appeared between 740 and 830 cm^{-1} could be attributed to C–H out-of plane bending vibration from aromatics and their derivatives disappeared gradually because of hydrogenation. These results indicated that the oxygenated groups (–OH, C=O, and C–O) were significantly removed in HDO oil.

In Fig. S4 (provided in supplementary data), the FT-IR spectra of the crude bio-oil and HDO oils at different hydrogen pressures are presented. The OH (carboxylic acids) stretching was decreased with the increase of pressure from 3.5 MPa to 7 MPa. CH (alkanes) aliphatic stretch increased considerably and CH bending stretch absorptions bands significantly decreased in this pressure range. The decrease in both OH and C=O stretch absorption bands and the increase in CH aliphatic stretch absorption band indicate that the carboxylic acids and other oxygenated chemical compounds were converted to hydrocarbons. At 5.2 MPa, the characteristic bands appeared between 740 and 830 cm^{-1} are ascribed to C–H out-of plane bending vibration from aromatics and their derivatives. The higher selectivity to the ring hydrogenation products at higher pressure is attributed to the increased H_2 availability (i.e., high H_2 surface coverage) at higher pressures [40]. At 7 MPa these bands disappeared. Increasing hydrogen pressure from 7 MPa to 8.6 MPa and 10 MPa did not have significant effect on FT-IR spectra of HDO oils. The results of FT-IR are in good agreement with the physico-chemical properties presented in Table S2 (supplementary data).

Fig. S5 (supplementary data) shows the FT-IR spectra of the crude bio-oil and HDO oils obtained at different reaction times of 15, 30, 60, and 90 min. According to this figure, it is clear that the very broad OH stretching absorption peak present in the raw bio-oil completely disappears when the reaction time increased from 15 to 30 min. The CH aliphatic stretch absorption band at 2800–3050 cm^{-1} in the bio-oil dramatically increased at reaction time of 30 min. Also the C=O absorption band of carbonyl functional group and the C–O absorption band of ether and alcohol functional groups disappeared during 30 min HDO. The changes in FT-IR spectrum of HDO oil absorption bands indicate that after 30 min reaction time at 450 °C in presence of 20% catalyst the carboxylic acids, aldehydes, and other oxygenated chemical compounds were converted to hydrocarbons. However the increase in reaction time from 30 min to 60 and 90 min did not have any significant effect on FT-IR spectrum of HDO oil. As previously discussed, after reaction time of 30 min, hydrocracking of organic components causes the organic liquid to convert to gas phase. The results of FT-IR at different reaction time were in good agreement with the properties of HDO oil (Table S4).

The functional groups present in the crude bio-oil and HDO oils were characterized by semiquantitative integration of ^{13}C NMR spectra. For comparison, a ^{13}C NMR experiment was carried out on commercial gasoline as well. The ^{13}C NMR spectra of the raw bio-oil, two HDO oils obtained at different reaction temperature (400 °C and 450 °C), and commercial gasoline (regular grade) are shown in Fig. S6 (supplementary data). The chemical shifts in the

spectrum were assigned to different functional groups [50,51]. The semiquantitative analysis of the ^{13}C NMR functional groups are presented in Table 4. The crude bio-oil had high amounts of carbohydrate degradation products, alcohols, ethers, methoxylated phenols, carboxylic groups, aldehydes, and ketones.

The characteristic peaks at the chemical shift of 0–28 ppm are assigned to saturated aliphatic carbon atoms. Clearly saturated aliphatics increased during HDO that could be due to hydrogenation of unsaturated aliphatics, HDO of alcohols, ethers, carboxylic acids, aldehydes, and ketones. Increasing the reaction temperature from 400 °C to 450 °C resulted in further increase in saturated aliphatic carbon atoms from 42.9% to 61.2% (Table 4). This would suggest that unsaturated compounds hydrogenate to saturated aliphatics.

The chemical shifts of 28–55 ppm are generally assigned to the unsaturated carbon atoms that are separated from oxygen atoms by at least two bonds [52]. The chemical shifts of 55–95 ppm are assigned to aliphatic carbon atoms attached to oxygen atoms in alcohols, esters, and anhydrous carbohydrates. After HDO process; alcohols, esters, and anhydrous carbohydrates fully hydrogenated, deoxygenated to aliphatics (at both reaction temperatures of 400 °C and 450 °C).

The characteristic peaks at chemical shift of 95–165 ppm are assigned to aromatic carbon atoms, phenolic compounds, and furans. These carbon atoms decreased after HDO at 400 °C. With the increase in reaction temperature from 400 °C to 450 °C, unsaturated aliphatics increased from 16.7% to 38.8% and aromatic compounds disappeared. At 450 °C both saturated and unsaturated aliphatic increased from 42.9% and 16.7% (at 400 °C) to 61.2% and 38.8%. These results indicate that hydrogenation of unsaturated aliphatics took place at 450 °C causing an increase in saturated and unsaturated aliphatics. However hydrogenation of unsaturated aliphatics did not fully taken place. Longer reaction times might result in complete hydrogenation of unsaturated aliphatic carbon atoms, however, as previously mentioned, longer reaction times can cause hydrocracking of organic compound to gases that reduces the organic liquid yield.

The chemical shifts of 165–180 ppm are generally assigned to carboxylic acids and esters. After HDO process, these compounds were completely hydrodeoxygenated. This can explain the increase in aliphatic hydrocarbons during HDO process (Table 4).

Characteristic peaks positioned at the chemical shift of 180–215 ppm are assigned to carbon atoms of carbonyl groups contained in aldehydes and ketones whose content decreased from 1.1% in bio-oil to zero in HDO oil at 400 °C and 450 °C indicating that aldehydes and ketones fully deoxygenated to aliphatic hydrocarbons. Interestingly, HDO oil obtained at 400 °C had a very similar ^{13}C NMR spectra to that of commercial gasoline (Fig S6 b and c, and Table 4).

Comparing to other bio-oils obtained from other feedstock, the catalytic PJ bio-oil used in this study had significantly less reactive groups of aldehydes, ketones, phenolics, and carboxylic acid fractions that are responsible for instability of bio-oil [51,52]. This could

Table 4
Functional group distribution of crude bio-oil, commercial gasoline, and HDO oils from ^{13}C NMR spectral integration.

Chemical shift region (ppm)	Dominant type of carbon	Percentage of carbon based on ^{13}C NMR analysis			
		Feed bio-oil	HDO oil-400 °C	Commercial gasoline	HDO oil-450 °C
0–28	saturated aliphatic groups	16.9	42.9	48.3	61.2
28–55	unsaturated aliphatic groups	10.6	16.7	13.3	38.8
55–95	alcohols, ethers, phenolic methoxys, anhydrosugars	13.1	0	1.7	0
95–165	aromatics, furans	56.7	40.4	36.7	0
165–180	organic (carboxylic) acids, esters	1.6	0	0	0
180–215	ketones, aldehydes	1.1	0	0	0

explain the accomplishment of HDO process in a single step.

No catalytic activity was seen in the case of blank experiments and the bio-oil polymerized (solidified) due to the presence of highly reactive compounds in bio-oil such as sugar derivatives and phenolic compounds [39,53]. The reactor was not catalytically active and did not influence the experiment.

3.4. Evaluation of HDO

To accurately visualize and extrapolate trends from elemental analysis, the elemental composition was plotted in Van Krevelen-type diagrams. The molar O/C versus molar H/C ratios of the crude bio-oil and the HDO oils are illustrated in Figs. 5–7 at different temperatures, initial hydrogen pressures, and reaction times respectively. Van-Krevelen diagrams drawn using elemental composition data reported in Tables 3, S2, and S4. These diagrams provide insight into the influences that process conditions have on the elemental composition of products [54]. According to Fig. 5, the O/C ratio of the oil decreased as a function of reaction temperature indicating deep deoxygenation occurs at higher temperatures. An increase in HDO process temperature resulted in an increase in H/C ratio from 1.81 at 350 °C to 2.36 at 450 °C. Fig. 5 shows that the ratio of H/C increased at lower temperatures while at higher temperatures O/C ratio decreased faster. This result suggests that hydrogenation reactions such as saturation of double bonds or alcohol formation from aldehydes and ketones may dominate at lower temperatures while HDO and dehydration reactions may be favored afterwards [45,47]. This result was in a good agreement with ^{13}C NMR results that were discussed in section 3.3 (Table 4).

HDO process using initial hydrogen pressure of 3.5 MPa decreased the O/C ratio from 0.29 of original bio-oil to 0.19 of HDO oil. The H/C increased significantly from 1.29 of bio-oil to 1.73 of HDO oil. Increasing the hydrogen initial pressure from 3.5 MPa to 5.2 MPa caused a slight decrease in O/C ratio from 0.19 to 0.13 while H/C increased considerably from 1.73 to 2.19. With the increase in hydrogen pressure from 5.2 MPa to 7 MPa O/C ratio dropped significantly from 0.13 to 0.00 while H/C ratio slightly increased from 2.19 to 2.36. These results indicate that hydrogenation reactions take place at lower pressure since the increase in H/C ratio were more notable than the decrease in O/C. However as shown in Fig. 6, deoxygenation of chemical components require higher hydrogen pressures. Increasing hydrogen initial pressure from 7 MPa to 8.6 MPa and 10 MPa did not have a significant effect on H/C and O/C ratios suggesting that 7 MPa of initial hydrogen pressure was sufficient to accomplish a complete HDO process.

As shown in Fig. 7, at reaction time of 15 min, the H/C ratio increased significantly to 2.03 compared to 1.29 of the original bio-

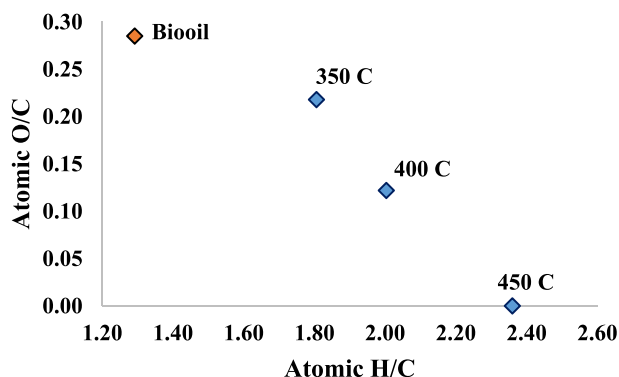


Fig. 5. Van Krevelen diagram of the HDO oil products at different reaction temperatures. H_2 initial pressure, and reaction time were 7 MPa and 60 min respectively.

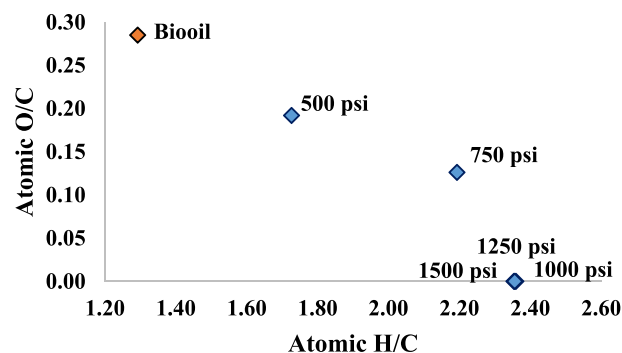


Fig. 6. Van Krevelen diagram of the HDO oil products at different H_2 initial pressures. Temperature, and reaction time were 450 °C, and 60 min respectively.

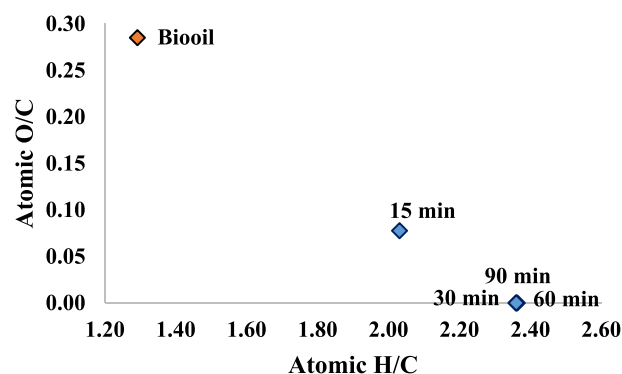


Fig. 7. Van Krevelen diagram of the HDO oil products at different reaction times. Temperature, and H_2 initial pressures were 450 °C, and 7 MPa respectively.

oil. Also the O/C ratio decreased from 0.29 of bio-oil to 0.08 of HDO oil. With the increase in reaction time from 15 min to 30 min H/C and O/C ratios reached to 2.36 and 0.00 respectively. This result suggested that at 450 °C with catalyst loading of 20% and initial H_2 pressure of 7 MPa, HDO reactions took place rapidly in the first 15 min and slowed down in the second 15 min. At the reaction time of 60 and 90 min the H/C ratio remained constant at the value of 2.36. The O/C ratio was 0.00 at reaction times of 60 and 90 min. This can be explained by completion of HDO reactions in 30 min and further hydrocracking of components in organic liquid to gas phase at higher reaction times. These results were in a good agreement with yield distribution and FT-IR results in sections 3.2.3 and 3.3 respectively.

4. Conclusion

A parametric hydrodeoxygenation study using commercial Ni/SiO₂–Al₂O₃ catalyst was carried out on pyrolysis oil derived from Pinyon Juniper. HDO experiments resulted in conversion of the bio-oil into two liquid phases (aqueous and organic), as well as gas and coke. The mass balance analysis of HDO experiments demonstrated that product yield distributions were considerably influenced by temperature, hydrogen pressure, and reaction time. High temperature lead to high catalytic activity for HDO. Hydration reactions mainly occurred at low temperature, while deoxygenation or cracking of oxygenated compounds were predominant at high temperature. With increasing reaction temperature from 350 °C to 450 °C the yield of upgraded oil decreased, but it was beneficial in improving the physicochemical properties. At low hydrogen pressure, hydrogenation dominated deoxygenation reactions. With increase in hydrogen initial pressure, deoxygenation reactions were

avored, in addition to hydrogenation. The increase in reaction time led to the decrease in organic liquid yield, due to the formation of gas-phase components and coke through cracking and polymerization respectively. The properties of HDO oils were significantly improved compared to the original bio-oil. This work provided a new alternative for production of liquid hydrocarbon from bio-oil.

Carbon loss due to coke and gas formation was a major challenge that reduced the yield of organic liquid product in HDO process. High amounts of hydrogen consumption during HDO operation was also another issue that adds to the cost of operation. Thus, it will be necessary to conduct a full economic optimization that incorporates the overall efficiency from pyrolysis to HDO, including the feedstock losses during pyrolysis. Future work should also investigate the rates of catalyst deactivation and coke formation, which could significantly alter the effectiveness of the process.

Acknowledgement

This research was supported by Utah Science Technology and Research (USTAR). The authors would like to acknowledge Dr. Sedat H. Beis for reviewing the manuscript. We would also thank Kyle Christian and Brandon Sargent for supplying the PJ oil.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.energy.2017.11.149>.

References

- [1] Sorrel S, Speirs J, Rentley R, Brandt A, Miller R. Global oil depletion: a review of the evidence. *Energy Pol* 2010;38:5290–5.
- [2] Zhang S, Yan Y, Li T, Ren Z. Upgrading of liquid fuel from the pyrolysis of biomass. *Biores Technol* 2005;96:545–50.
- [3] Lee JW, Hawkins B, Day DM, Reicosky DC. Sustainability: the capacity of smokeless biomass pyrolysis for energy production, global carbon capture and sequestration. *Energy Environ Sci* 2010;3:1695–705.
- [4] Yathavan BK, Agblevor FA. Catalytic pyrolysis of pinyon juniper using red mud and HZSM-5. *Energy Fuels* 2013;27:6858–65.
- [5] Mullen CA, Boateng AA. Production and analysis of fast pyrolysis oils from proteinaceous biomass. *Bioenergy Res* 2011;4:303–11.
- [6] Agblevor FA, Beis SH, Mante OD, Abdoulmoumine N. Fractional catalytic pyrolysis of hybrid poplar wood. *Ind Eng Chem Res* 2010;49:3533–8.
- [7] Boateng AA, Dugaard DE, Goldberg NM, Hicks KB. Bench-scale fluidized-bed pyrolysis of switchgrass for bio-oil production. *Ind Eng Chem Res* 2007;46:1891–7.
- [8] Schnitzler MI, Monreal CM, Facey GA, Fransham PBJ. The conversion of chicken manure to biooil by fast pyrolysis I. Analyses of chicken manure, biooils and char by ^{13}C and ^1H NMR and FTIR spectrophotometry. *J Environ Sci Health B* 2007;42:71–7.
- [9] Agblevor FA, Mante OD, McClung R, Oyama ST. Co-processing of standard gas-oil and biocrude oil to hydrocarbon fuels. *Biomass Bioenergy* 2012;45:130–7.
- [10] Kim TS, Sh Oh, Kim JY, Choi IG, Choi IW. Study on the hydrodeoxygenative upgrading of crude bio-oil produced from woody biomass by fast pyrolysis. *Energy* 2014;68:437–43.
- [11] Mortensen PM, Grunwaldt JD, Jensen PA, Knudsen KG, Jensen AD. A review of catalytic upgrading of bio-oil to engine fuels. *Appl Catal A General* 2011;407:1–19.
- [12] Bridgwater AV, Czernik S, Piskorz J. Progress in thermochemical biomass conversion. *Blackwell Wissenschafts-Verlag*; 2001. p. 977–97.
- [13] Oasmaa A, Elliott DC, Korhonen J. Acidity of biomass fast pyrolysis oils: state of the art for the end users. *Energy Fuels* 2010;24:6548–54.
- [14] Mante OD, Agblevor FA. Storage stability of biocrude oils from fast pyrolysis of poultry litter. *Waste Manage* 2012;32:67–76.
- [15] Oasmaa A, Kuoppala E. Fast pyrolysis of forestry residue: 3. Storage stability of liquid. *Energy Fuels* 2003;17:1075–84.
- [16] Boucher ME, Chaala A, Pakdel H, Roy C. Bio-oils obtained by vacuum pyrolysis of softwood bark as a liquid fuel for gas turbines. Part II: stability and ageing of bio-oil and its blends with methanol and a pyrolytic aqueous phase. *Biomass Bioenergy* 2000;19:351–61.
- [17] Sh Ahmadi, Reyhanitash E, Yuan Z, Rohani S. Upgrading of fast pyrolysis oil via catalytic hydrodeoxygenation: effects of type of solvents. *Renew Energy* 2017;114:376–82.
- [18] Zhang Z, Bi P, Jiang P, Fan M, Sh Deng, Zhai Q, et al. Production of gasoline fraction from bio-oil under atmospheric conditions by an integrated catalytic transformation process. *Energy* 2015;90:1922–30.
- [19] Lee H, Kim Y, Lee I, Joen J, Jung S, Chung JD, et al. Recent advances in the catalytic hydrodeoxygenation of bio-oil. *Korean J Chem Eng* 2016;33:3299–315.
- [20] Sh Xiu, Shahbazi A. Bio-oil production and upgrading research: a review. *Renew Sustain Energy Rev* 2012;16:4406–14.
- [21] Choudhary TV, Phillips CB. Renewable fuels via catalytic hydrodeoxygenation. *Appl Catal A General* 2011;397:1–12.
- [22] Yang Y, Lv G, Deng L, Lu B, Li J, Zhang J, et al. Renewable aromatic production through hydrodeoxygenation of model bio-oil over mesoporous Ni/SBA-15 and Co/SBA-15. *Microporous Mesoporous Mater* 2017;250:47–54.
- [23] Joshi N, Lawal A. Hydrodeoxygenation of pyrolysis oil in a microreactor. *Chem Eng Sci* 2012;74:1–8.
- [24] He Z, Wang X. Hydrodeoxygenation of model compounds and catalytic systems for pyrolysis bio-oils upgrading. *Catal Sustain Energy* 2013;28–52.
- [25] Elkasabi Y, Mullen CA, Pighinelli LMT, Boateng AA. Hydrodeoxygenation of fast pyrolysis oils from various feedstocks using carbon-supported catalysts. *Fuel Process Technol* 2014;123:11–8.
- [26] Oh SH, Hwang H, Choi HS, Choi JW. The effect of noble metal catalysts on the bio-oil quality using the hydrodeoxygenative upgrading process. *Fuel* 2015;153:535–43.
- [27] Han Y, McIlroy DN, McDonald AG. Hydrodeoxygenation of pyrolysis oil for hydrocarbon production using nanospring based catalysts. *J Anal Appl Pyrolysis* 2016;117:94–105.
- [28] Elkasabi Y, Liu Q, Choi YS, Strahan G, Boateng AA, Regalbutto JR. Bio-oil hydrodeoxygenation catalysts produced using strong electrostatic adsorption. *Fuel* 2017;207:510–21.
- [29] Li X, Chen G, Liu C, Ma W, Yan B, Zhang J. Hydrodeoxygenation of lignin-derived bio-oil using molecular sieves supported metal catalysts: a critical review. *Ren Sustain Energy Rev* 2017;71:296–308.
- [30] Perlak RD, Stokes BJ. U.S. Department of Energy. Oak Ridge: Oak Ridge National Laboratory; 2011. TN. 227 p. ORNL/TM-2011/224.
- [31] Miller RF, Bates JD, Svecar TJ, Pierson FB, Eddleman LE. Technical bulletin 152. Corvallis, OR: Corvallis Agricultural Experiment Station, Oregon State University; 2005.
- [32] Carrara PE, Carroll TR. The determination of erosion rates from exposed tree roots in the Piceance basin, Colorado. *Earth Surf Process* 1979;4:307–17.
- [33] Ansley RJ, Wiedemann HT, Castellano MJ, Slosser JE. Herbaceous restoration of juniper dominated grasslands with chaining and fire. *Rangel Ecol Manage* 2006;59:171–8.
- [34] Baughman C, Forbis TA, Provencher L. Response of two sagebrush sites to low-disturbance, mechanical removal of Piñon and Juniper. *Invasive Plant Sci Manage* 2010;3:122–9.
- [35] Lee EH, Park R, Kim H, Park SH, Jung S, Jeon J, et al. Hydrodeoxygenation of guaiacol over Pt loaded zeolitic materials. *J Ind Eng Chem* 2016;37:18–21.
- [36] Mante OD, Agblevor FA. Catalytic pyrolysis for the production of refinery-ready biocrude oils from six different biomass sources. *Green Chem* 2014;16:3364–73.
- [37] ASTM Standard E1756-08. In 2008 annual book of ASTM standards, 11.05. West Conshohocken, PA: American Society for Testing and Materials; 2008.
- [38] ASTM Standard E1755-01. In 2008 annual book of ASTM standards, 11.05. West Conshohocken, PA: American Society for Testing and Materials; 2008.
- [39] Hu X, Wang Y, Mourant D, Gunaman R, Lievens C, Chaiwat W, et al. Polymerization on heating up of bio-oil: a model compound study. *AIChE* 2013;59:888–900.
- [40] Tanneru SK, Steele PH. Direct hydrocracking of oxidized bio-oil to hydrocarbons. *Fuel* 2015;154:268–74.
- [41] Danaci S, Protasova L, Lefever J, Bedel L, Guiler R. Efficient CO_2 methanation over $\text{Ni}/\text{Al}_2\text{O}_3$ coated structured catalyst. *Catal Today* 2016;273:234–43.
- [42] Wan H, Chaudhari RV, Subramaniam B. Catalytic Hydroprocessing of p-cresol: metal, solvent and mass transfer effect. *Top Catal* 2012;55:129–39.
- [43] Asphaug S. Catalytic hydrodeoxygenation of bio-oils with supported MoP-catalysts. *TKP 4900 Thesis*. 2013. p. 52–4.
- [44] Shin EJ, Keane MA. Gas-phase hydrogenation/hydrogenolysis of phenol over supported nickel catalysts. *Ind Eng Chem Res* 2000;39:883–92.
- [45] Gutierrez A, Kaila RK, Honkela ML, Slioor R, Krause AOL. Hydrodeoxygenation of guaiacol on noble metal catalysts. *Catal Today* 2009;147:239–46.
- [46] Elliott DC, Hart TR, Neuenschwander GG, Rotness LJ, Zacher AH. Catalytic hydroprocessing of biomass fast pyrolysis bio-oil to produce hydrocarbon products. *Environ Prog Sustain Energy* 2009;28:441–9.
- [47] Zhu Z, Rosendahl L, Toor SS, Yu D, Chen G. Hydrothermal liquefaction of barley straw to bio-crude oil: effect of reaction temperature and aqueous phase recirculation. *Appl Energy* 2015;137:183–92.
- [48] Garcia T, Veses A, Lopez JM, Puertolas B, Perez-Ramirez J, Callen MS. Determining bio-oil composition via chemometric tools based on infrared spectroscopy. *Sus Chem Eng* 2017;5:8710–9.
- [49] Durak H, Aysu T. Effects of catalysts and solvents on liquefaction of *Onopordum heteranthum* for production of bio-oils. *Bioresour Technol* 2014;166:309–17.
- [50] Mante OD, Agblevor FA, Oyama ST, McClung R. The effect of hydrothermal treatment of FCC catalysts and ZSM-5 additives in catalytic conversion of biomass. *Appl Catal A General* 2012;445:312–20.
- [51] Zhang X, Chen L, Kong W, Wang T, Zhang Q, Long J. Upgrading of bio-oil to boiler fuel by catalytic hydrotreatment and esterification in an efficient process. *Energy* 2015;84:83–90.

- [52] Strahan GD, Mullen CA, Boateng AA. Characterizing biomass fast pyrolysis oils by ^{13}C NMR and chemometric analysis. *Energy Fuels* 2011;25:5452–61.
- [53] Kadarwati S, Oudenhoven S, Schagen M, Hu X, Garcia-Perez M, Kersten S, et al. Polymerization and cracking during the hydrotreatment of bio-oil and heavy fractions obtained by fractional condensation using Ru/C and NiMo/Al₂O₃ catalyst. *J Anal Appl Pyrolysis* 2016;118:136–43.
- [54] Van Krevelen D. Graphical-statistical method for the study of structure and reaction process of coal. *Fuel* 1950;29:269–84.